A Bite into the Data:

A Comprehensive Deep Dive into a Healthier Snacking Business

Mid-Term Report

Submitted by

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Contents

1 Executive Summary and Title

This analysis evaluates the sales efficiency of Poshtique, a Mumbai-based healthy snack company founded by Purnima Mehta and Noopur Mehta. Despite operating in a growing industry, Poshtique is currently experiencing several sales and marketing challenges that are negatively impacting profitability. Some of these hardships elucidated upon in this analysis include forecasting seasonal demand, difficulties in stock management and the limited understanding of customer patterns. This analysis aims to provide practical recommendations for the addressing these challenges through data-driven insights.

To facilitate qualitative and quantitative analysis, a year's worth of sales data in the form of CSV files were provided - Summary and Details. The summary file contains key order information, including customer name, address, grand total, shipping and GST. The Orders file, on the other hand, provides granular item-level details, including product name, quantity, unit price and stock remaining. Both files are relationally linked, allowing a more in depth analysis of sales performance. Together, these datasets offer a foundation for deriving implementable findings.

Business context and operational challenges were gathered through exhaustive interviews with the company's owners. The raw data was subsequently cleaned and enriched (e.g., adding geolocation fields) to support deeper analysis. Descriptive statistics and diagnostic modelling were conducted to uncover key patterns, such as monthly product sales, spending distribution, and customer retention metrics.

2 Proof of Data Originality

Company Website – https://www.poshtique.in/
Founder Interaction Video – Link

NOC Signed Letter

Poshtique™

A venture by **Health Harmony**

20th May 2025

No Objection Certificate

To Whomsoever Concerned,

I, Noopur Mehta, founder of Poshtique, hereby certify that Mr. Henil Diwan has taken data from our business as a part of his Business Data Management Capstone Project at IIT Madras.

We give consent to Mr. Henil Diwan to make use of the provided data for his project. This No Objection Certificate (NOC) is issued at the request of Mr. Henil Diwan for the specific purpose mentioned above.

Sincerely,

Noopur K Mehta

Noopur Mehta

Poshtique

522, Prestige Industrial Estate, Baudi Lane, Off Marve Road, Malad West, Mumbai – 400064
Contact no.: 9833433687/9820168684 Email id: poshtique.in@gmail.com

Poshtique Website

Pictures 2.2 – Screenshot of Poshtique's Products

3 Metadata and Descriptive Statistics

3.1 Metadata

Raw Data - Summary & Details

Note: If you encounter issues opening the file in the Chrome browser, please try copying and pasting the link directly into the address bar.

For business analytical purposes, Poshtique provided me with transactional data of 340 unique customers for the calendar year 2024. The data was provided in comma separated value (CSV) format, organized in two distinct files – Summary and Details. As the company is based and operates in the Indian market, all monetary values are denominated in Indian Rupees (INR). This dataset serves as a detailed account of Poshtique's sales throughout the year.

File 1: Summary – A high-level overview of orders is provided in the Summary file, with each row representing an individual order. The file contains details such as customer names and addresses, order values, discounts offered and gst collected. This file, structured as a master record, supports order-level analysis of Poshtique's transactions.

- Number of Rows: 544 (each representing a unique order)
- Number of Columns: 15

Fields Description –

Name	Data Type	Description
Order No	String	Unique identifier for each order
Order Date	Date	Date when the order was placed
Customer ID	String	Unique identifier for each customer
Customer Name	String	Name of the customer
Area	String	Local neighborhood of the customer
City	String	City of the customer
State	String	State of the customer
Pincode	Number	Postal Code of the customer's address
Order Total	Float	Total value of items before discounts, taxes and shipping
Discount (%)	Number	Percentage discount offered for the order
Discount Amount	Float	Absolute discount value in INR
GST	Float	Goods and Service Tax collected
Shipping	Float	Shipping charge applied
Grand Total	Float	Final amount paid after applying all adjustments
Channel	String	Sales Channel through which the order was placed (Online or Amazon)

Sheet 2: Details – Product-level information of orders is provided in Details file. Each row in the file corresponds to an individual product within an order. This provides more granular details such as product-specific performance, quantity ordered and inventory levels. This file is designed to be linked to the Summary file, enabling in-depth analysis and insights.

- Number of Rows: 884
- Number of Columns: 7

Fields Description –

Name	Data Type	Description
Order No	String	The order number corresponding to the particular order containing the product. Can function as a foreign key linking back to the summary sheet
Product Name	String	Name of the product ordered
Quantity Ordered	Number	Number of units ordered for the product
Unit Price	Float	Price per single unit of the product
Total Price	Float	Total value of all the units of the product (Quantity * Unit Price)
Stock Remaining	Number	Inventory of the product remaining after fulfilling the order
Replenishment	Number	Quantity of stock replenished on the date of the order (may be null if no replenishment occurred)

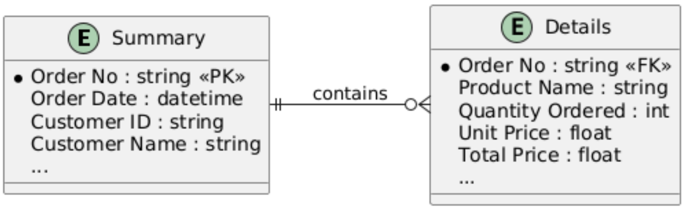

Figure 3.1.1 – Relational Schema of the Dataset

3.2 Descriptive Analysis

In the year 2024, Poshtique achieved gross sales amounting to ₹5,05,500, while net sales stood at ₹6,16,081.

Metric	Order Total	Discount (%)	Discount Amount	GST	Shipping	Grand Total
Count	544	544	544	544	544	544
Mean	961.95	5.07	48.92	115.43	104.03	1,132.50
Std Dev	679.64	9.61	122.94	81.56	2.89	727.80
Min	100.00	-	-	12.00	57.00	173.00
25th Percentile	450.00	-	-	54.00	102.50	562.13
Median (50th pct)	800.00	-	-	96.00	104.00	999.00
75th Percentile	1,300.00	10.00	21.25	156.00	105.63	1,505.50
Max	5,000.00	40.00	900.00	600.00	109.50	5,706.00

Table 3.2.1 – Measures of Central Tendency for Summary file

Table 3.2.1 describes measures of central tendency for the Summary file. The analysis of this data reveals key insights, such as –

- Orders values range from ₹173 to ₹5,706
- 25% of orders are under ₹562, 50% are under ₹999, and 75% of orders are under ₹1,505
- Over 50% of orders received no discount, with the highest discount offered being 40%
- The average shipping cost is approximately ₹104, with a high level of consistency and low variance.

Metric	Unit Price (₹)	Total Price (₹)	Stock Remaining	Replenishment Day
Count	884	884	884	32
Mean	211.37	571.83	47.68	83.56
Std Dev	679.64	9.61	122.94	81.56
Min	100.00	100.00	-	22.00
25th Percentile	150.00	250.00	19.75	57.75
Median (50th pct)	200.00	400.00	43.00	78.00
75th Percentile	250.00	762.50	70.25	106.25
Max	350.00	2,700.00	142.00	153.00

Table 3.2.2 – Measures of Central Tendency for Details file

Table 3.2.2 similarly describes the measures of central tendency for the Details file. From the data, the following conclusions can be drawn –

- The average cost of products is ₹211.37, with the least expensive item priced at ₹100 and the most expensive product priced at ₹350
- The business maintains approximately 48 units of stock for each item, while approximately 83 units are replenished on any given replenishment day

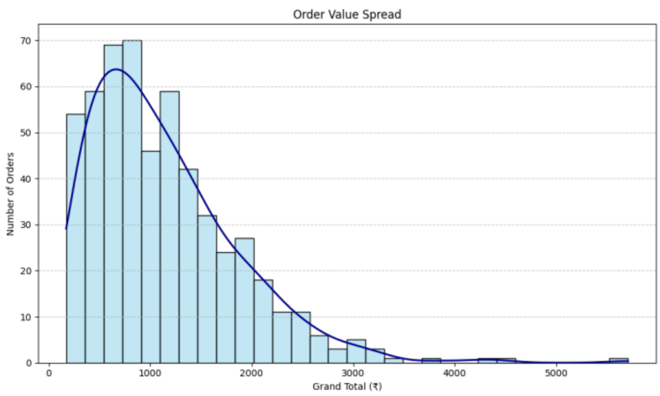

Figure 3.2.1 – Distribution of Grand Totals

Graph 3.2.1 illustrates the distribution of order values (Grand Total) in the form of a histogram. This indicates a right-skewed distribution: majority of the orders fall between ₹662–₹1,674. However a few large orders push the mean up. The long tail of high-value orders suggests a small number of customers make large purchases.

Channel	Total Revenue Gained	Number of Orders	Average Order Total	Max Order Total	Min Order Total	Percentage of Orders
Amazon	3,73,009.00	327	1,140.70	5,706.00	173.00	60.11%
Online	2,43,072.00	217	1,120.15	4,245.50	213.00	39.89%
Grand Total	6,16,081.00	544	1,132.50	5,706.00	173.00	100.00%

Table 3.2.3 – Distribution of Channels

Table 3.2.3 illustrates that Amazon is the dominant sales channel, generating 60% of orders and the highest revenue (₹3.73L) with a slightly higher average order value than the online store. This disparity may be attributed to higher traffic on Amazon.

Figure 3.2.2 – Distribution of Order Quantity

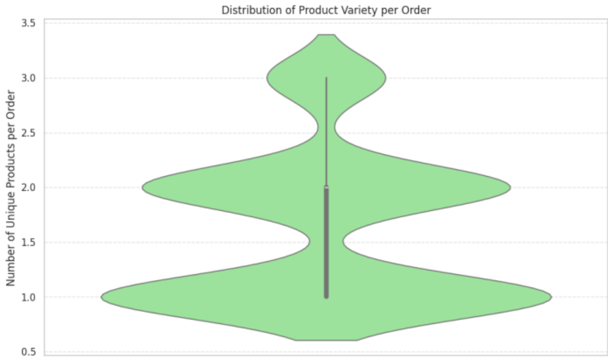

Figure 3.2.3 – Distribution of Order Variety

The violin graphs in figures 3.2.2 and 3.2.3 present the distributions of order quantity (i.e., total number of products ordered) and product variety (i.e., the number of distinct products ordered) per customer order, respectively. The data reveals that while the majority of customers order between 3 and 5 products per order, with some outliers placing orders of ten or more products, most customers typically select a only one or two types of products per order.

4 Analysis Process/Method

Various stages of the Data Analysis Life Cycle (DALC) were employed to facilitate a descriptive and diagnostic analysis of the data –

- Data Collection and Business Understanding –
  In-depth knowledge and comprehensive understanding of the intricacies of Poshtique as a business were gained through a series of extensive, one-on-one interviews conducted with the company's owners. These discussions proved to be highly informative, providing valuable insights into the ground reality and market-related challenges faced by the business. The owners additionally provided me with one year (2024) of sales data for analysis.

- Data Preprocessing –
  Thorough preprocessing steps were undertaken to enhance the reliability and comprehensiveness of the data analysis. Missing or null values were systematically addressed. This included manually updating entries where only pincode data or

incomplete state information was available in customer addresses. Additionally, Geocoding was performed using the Excel VLOOKUP function against a global dataset of latitudes and longitudes. Products were also categorized in broad categories of "sweet" and "savory" for further understanding preferences.

- Descriptive Analysis –
  Several statistical methods were utilized in the process of creating a descriptive analysis to ensure a thorough understanding of the data. Key statistical measures, such as mean and standard deviation, were computed using Excel's built-in formulas, while graphs and visualisations were created using Python (Scikit) and Power BI.
- Data Modelling –
  A preliminary diagnostic analysis was done to identify trends in the data and understand how they relate to the challenges Poshtique is facing. To identify these trends, visualizations such and funnel graphs and heatmaps were modelled, primarily developed using Power BI and Python libraries such as Scikit and Matplotlib. The visualisations were then systematically reviewed to extract critical insights such as product wise performance, seasonal demand trends and top customers.

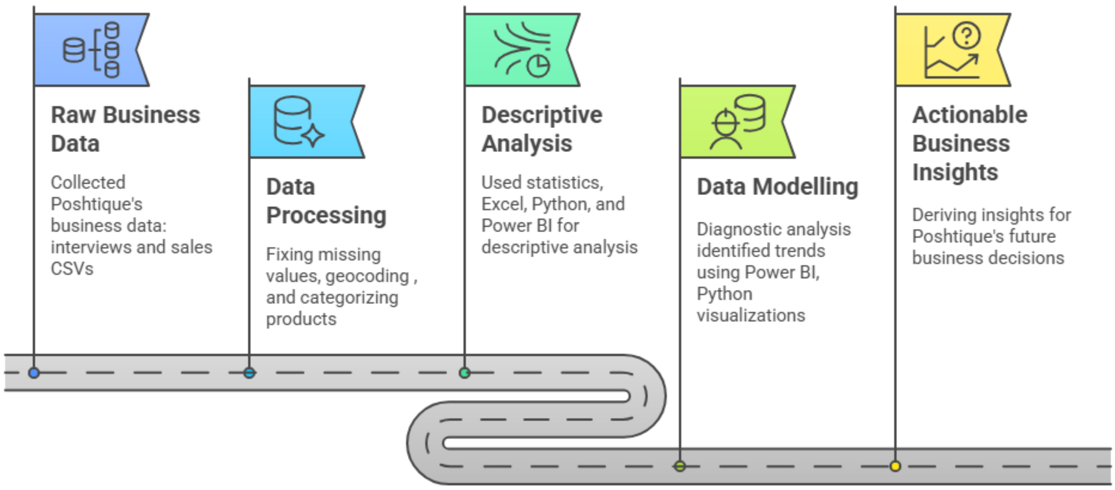

Figure 4.1 – Poshtique Data Analysis Process

5 Results and Findings

Based on the analysis, several key insights have been identified, highlighting significant trends and patterns that are crucial in understanding and addressing Poshtique's challenges.

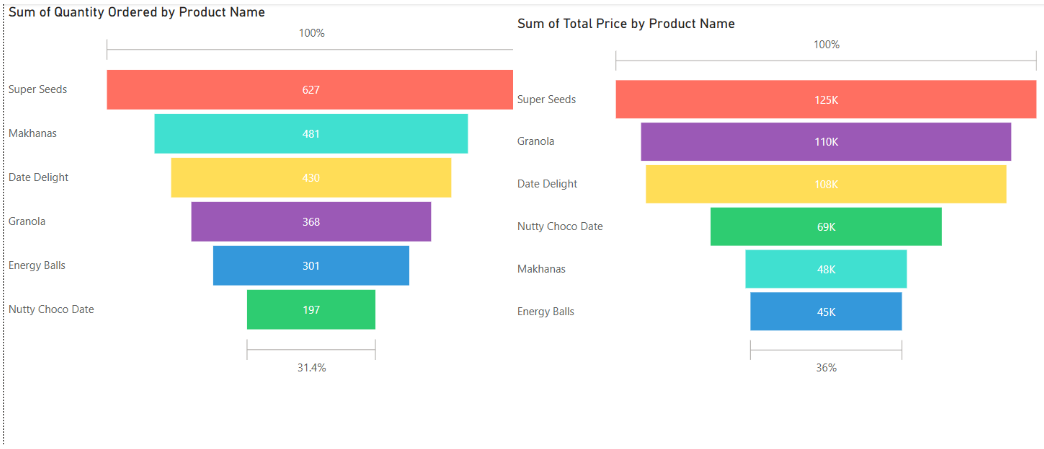

Figure 5.1 – Product-Wise Sales Comparison

Figure 5.1, the funnel graph, shown above provides product-wise revenue generated and quantity purchased. It can be inferred that while super seeds, makhanas and date delights account for the majority of the products ordered by quantity, granola surpasses makhanas in terms of revenue generated. This is primarily due to the lower unit price of makhanas. Therefore, care should be taken to stock higher quantities of these fast-moving products to prevent frequent understocking.

Figure 5.2 – Top 10 Customers By Revenue

Figure 5.2 shows the top 10 customers by revenue generated for Poshtique. Notably, Surya Kanta Kumari and Kaluram Rao have the highest order counts (17 and 16 orders respectively), but their revenue contributions are comparatively lower (₹8,342 and ₹8,559), indicating a lower average order value, while Gungun Sonkar and Toran Tiwari have made fewer purchases (7 and 6 orders respectively), yet still rank in the top 10, suggesting higher-value transactions. All of the top-revenue customers are repeat customers with multiple orders, underlining the importance of investing in strategies that foster long-term customer loyalty.

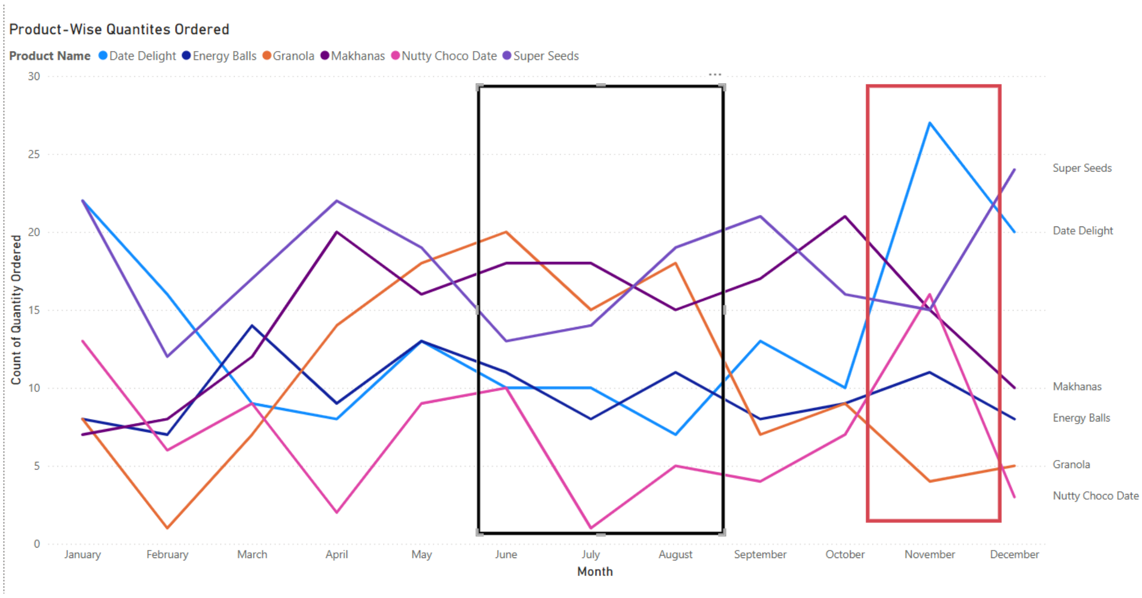

Figure 5.3 – Product Sales Over Time

Figure 5.3 illustrates trends in product sales through time. The region highlighted by the red box indicates an increased amount of sales, especially for sweeter products, which corresponds with the festival period (Diwali). The area within the black box represents a notable decline in sales. A further more detailed analysis is required to understand the underlying causes for this decrease in product sales.

Additionally, this graph reveals significant volatility in product demand, with customer preferences shifting dramatically from month to month. For example, Nutty Choco Date experiences sharp peaks and troughs, indicating it may be a promotional or occasion-specific product rather than one with stable year-round demand. In contrast, Super Seeds and Makhanas show more consistent, moderate fluctuations, suggesting they may be part of regular consumption habits.

The graph also reflects possible product life cycle stages: some items like Date Delight appear to gain popularity as the year progresses, potentially due to successful marketing or word-of-mouth, while others like Granola show mid-year peaks followed by a gradual decline, hinting at saturation or fading novelty. These patterns can guide product-specific strategies — for example, positioning Granola as a summer item or boosting visibility for rising products like Date Delight during early quarters.

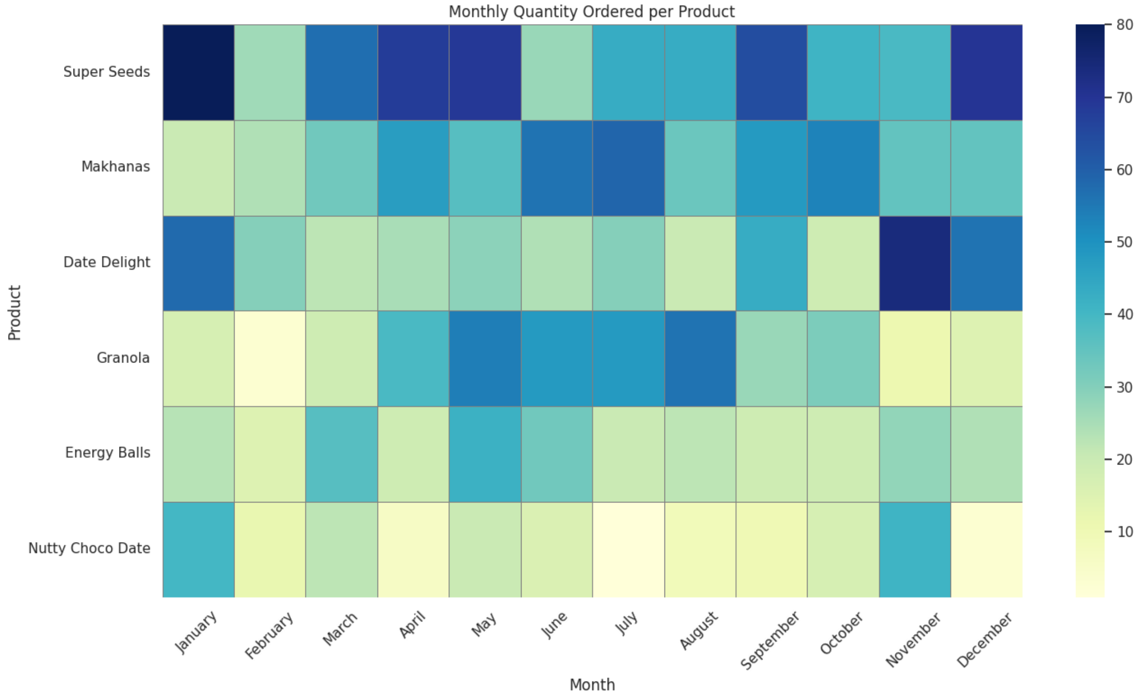

Figure 5.4 – Heatmap of Product Sales

Figure 5.4, the heatmap, focuses on relative intensities making inter-product comparisons easier. This helps in painting a clear picture of which products consistently outperform other products across various months. Super seeds, with its consistent darker shades, stands out as the top-seller, warranting inventory prioritization and premium placement. On the other hand, Energy Balls and Nutty Choco Dates occupy the lighter end of the spectrum for most of the year, indicating that they are underperforming products that may need repositioning.

6 Next Steps

The preliminary analysis presented in this Mid-Term Submission highlights significant patterns in the data, providing a solid foundation for a further refinement in the final report. The successful geocoding of latitudes and longitudes enables the creation of geotagged maps to understand customer distribution across India. Additionally, an understanding of the top and repeat customers allows for a deeper analysis of customer preferences, enabling precise categorisation. An examination of SKU popularity and seasonal demand trends offer the opportunity to calculate stock velocity and evaluate it against lead times.